



CM for ZTE Axon 7

Axon 7 could get official CyanogenMod ROM that ZTE and CM Team is developing.
<http://dgit.in/CMAXon7>



Vulnerable Tesla S

Tesla Model S yet to be completely secure against the remote vulnerable attacks.
<http://dgit.in/TesSHack>

Tomorrow's Tech

voice assistant that can sense emotion, converse, make decisions and on the back of a powerful ANN, actually learn from its conversations. Companies like Google are at the forefront of this new and exciting subset of AI.

Moving beyond voice and expression based systems even simple texts can sometimes give away emotions. For a company that centres its business around a search engine this means a big deal. Figuring out the actual context and intent of a search query has attracted much attention of late, especially since Google announced a transition to Hummingbird, triggering a buzz around what is now called as semantic search. All the above technologies, impressive as they may seem, don't draw up an intimidating figure as of now, only invoking an image of a drab computer in a dingy room going about its business. How far are we from humanoids that are an aggregation of the systems described above, possessing the ability to sense,



Pepper bot in a retail environment

understand and even replicate emotion? As far back in 2010, European scientists made a humanoid called Nao that could learn from human interaction, sense mood and even develop a personality of its own. Last year Softbank unveiled Pepper, claiming it to be the world's first emotionally intelligent robot. With the inclusion of big players, it is not long before we see improved variants that employ techniques that are much more

superior than the ones currently in use. The rise of machine that can recognise and respond to emotions however can have some troubling consequences too. How close are we before we create a sentient AI that goes all Terminator on us?

The difficulty in programming emotional machines

Luckily (or not, depending on which side of the debate you are on) current technology is far from replicating or recognizing human emotion in its true glory. Ultimately almost every machine learning system relies on finding patterns within the vast amount of data input that it perceives. Current machines are good at sensing pixel data or parsing text or sound information to find patterns within them. A machine can see a group of smiling pictures and correlate it to a happy group of people, but what happens when this correlation fails because one or more persons were faking emotion? Current models of emotion detection largely depends on sensing physiological markers for emotion, like a frown might correspond to dissatisfaction and an open mouth, disbelief. But human emotion is far too complex to lend itself to this treatment. A machine can detect a quickened pulse or heart beat and subsequently match it to a state of fear. But what if the heightened state is excitement or love?

A recent study asserts that when computers learn human languages, they also pick up the large range of biases that come with it. Emotion as a concept doesn't generalise easily due to which modelling emotion is a headache for scientists. Then there is the problem of replicating human emotion. It would seem that human emotion expression is simply coordinated muscle movement. But even that muscle movement is a difficult act to pull off in a machine. What's more, machines also need to circumvent the 'uncanny valley', a sense of discomfort that settles in humans when they see 'human like' behaviour in an obvious non human.

Last but not the least if we truly need affective computing to trickle down to

our daily lives it would mean an overhaul in terms of hardware optimization. Machine learning algorithms are power and memory intensive and implementing deep learning tools requires massive computing power. Our chips need to consume less power, store more in less space and work faster. .



A human like robot (Repliee) usually triggers high neuron activity due to a phenomena called the uncanny valley

AI's have always occupied the fancy of writers and scientists alike. In some form or another, AI's, even emotional AI's have appeared in pop culture for quite sometime. It comes as no wonder that a robot that we could talk to, like we would converse with our human friends, sounds like a natural extension of normal robots and in general seems like a good idea. But making a machine so close to a human that it even feels like one, can have repercussions that are chilling to even think about. If a machine can anticipate our actions it might even influence it. And, if we grant an inanimate object the power to animate, we are giving birth to a whole new species, an intelligent one at that. An emotionally intelligent AI might very well be great for us. But if literature and film mirror our future, we may be better off without one riddled with emotionally intelligent machines. **[1]**